

1. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following.

- a. A red card

$$\frac{n(\text{red cards})}{n(\text{cards})} = \frac{26}{52} = \frac{1}{2}$$

- b. A queen

$$\frac{n(\text{queens})}{n(\text{cards})} = \frac{4}{52} = \frac{1}{13}$$

- c. A club

$$\frac{n(\text{clubs})}{n(\text{cards})} = \frac{13}{52} = \frac{1}{4}$$

- d. The queen of clubs

$$\frac{n(\text{queen of clubs})}{n(\text{cards})} = \frac{1}{52}$$

- e. A queen or a club

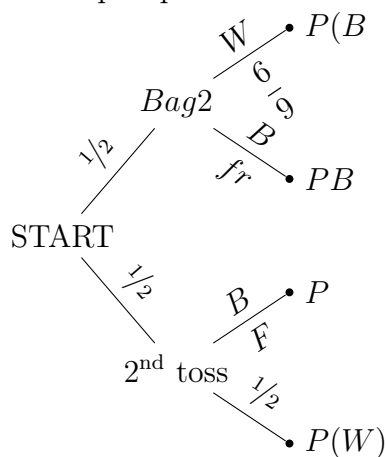
$$\frac{n(\text{queens or clubs})}{n(\text{cards})} = \frac{4 + 13 - 1}{52} = \frac{16}{52} = \frac{4}{13}$$

- f. not a queen

$$1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}$$

2. Three coins are tossed. Find each of the following:

- a. the sample space



- b.

- c. the event,  $E_1$ , that exactly two are tails

- d. the event,  $E_2$ , that at least two are tails.

- e. the probability and odds of  $E_1$

- f. the probability and odds of  $E_2$

3. Five cards are dealt (without replacement) from a well-shuffled standard deck. **Just write the formula** for the probability of the following:

- a. being dealt 5 spades

- c. being dealt all the same suit.

- b. being dealt 4 spades and 1 heart

- d. not being dealt all the same suit.

4. A drawer has 7 socks. 4 socks are black and 3 are white socks. John randomly pulls out 4 socks. Find the probability of the following:

- a. all 4 socks are black.
- b. exactly 2 are white.
- c. at least 3 are white.
- d. at most 2 are black.

5. Gregor's Garden Corner buys 40% of its plants from the Green Growery and the rest from Herb's Herbs. Of the plants from the Green Growery, 20% must be returned, and 10% of those from Herb's Herbs must be returned.

- a. draw a tree diagram describing the probability
- b. find the probability that a plant must be returned and it is from Herb's Herbs.
- c. find the probability that a plant from the Green Growery must be returned.
- d. find the probability that a plant must be returned.

6. Use the provided table to answer the following questions.

| Age   | Obama | McCain | Other/NA |
|-------|-------|--------|----------|
| 18-29 | 11.9% | 5.8%   | 0.4%     |
| 30-44 | 15.1% | 13.3%  | 0.6%     |
| 45-64 | 18.5% | 18.1%  | 0.4%     |
| 65+   | 7.2%  | 8.5%   | 0.3%     |

- a. What is the probability that a voter under 45 voted for McCain.
- b. What is the probability that a person who voted for McCain is 45 or older.

7. MHU is holding a raffle to raise money for a new internet service. The tickets cost \$10 each. The value and number of each prize to be given away is listed in the provided table. Find the expected value of a ticket under the given circumstances.

| Prize    | Value   | #   |
|----------|---------|-----|
| Vacation | \$3,000 | 1   |
| Stereo   | \$500   | 5   |
| Books    | \$100   | 10  |
| T-shirt  | \$20    | 100 |

a. 200 tickets

b. 1000 tickets

8. Of all the flashlights in a large shipment, 15% have a defective bulb, 10% have a defective battery, and 5% have both defects. If you buy one of these flashlights, find the probability of the following:
- a. a defective bulb or a defective battery
  - b. a good bulb or a good battery
  - c. a defective bulb and a defective battery.
  - d. just a defective bulb
9. A single card is dealt from a well-shuffled standard deck. The probability of being dealt a jack is  $\frac{4}{52}$ , and the probability of being dealt a red card is  $\frac{1}{2}$ .
- a. Are the events “being dealt a jack” and “being dealt a red card” independent?
  - b. Are the events “being dealt a jack” and “being dealt a red card” mutually exclusive?
  - c. Are the events “being dealt a jack” and “being dealt a queen” mutually exclusive?
  - d. Suppose instead that a card is dealt, replaced, and then another card is dealt. Are the events “being dealt a jack 1<sup>st</sup>” and “being dealt a queen 2<sup>nd</sup>” independent? are they mutually exclusive?
10. A fair six-sided dice is rolled 5 times. Find the probability of the following:
- a. All 2's are rolled.
  - b. Four 2's are rolled and the 5<sup>th</sup> roll is a 1.